HCL PROJECT-BASED LEARNING | CYBERSECURITY

Security Incident Detection & Analysis Report

Incident Investigation, Root-Cause Analysis, & Remediation Playbook

Course / Module	Security Incident Detection and Analysis (Project 1)
Project Scope	DNS Tunneling, Covert C2 Communication & Data Exfiltration
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Live Deployment & Repo	Web App: incident-detection-dns-tunneling.streamlit.app GitHub: github.com/Harsh18062005/incident-detection-dns-tunneling

Section a. Incident Summary

Incident Type	DNS Tunneling / Data Exfiltration via C2 Channel (MITRE ATT&CK T1071.004, T1041)
Date & Time of Incident	2026-10-31 00:20:00 UTC (Continuous SOC Monitoring Session)
Severity Level	**CRITICAL (Risk Score 7/7 - Multi-Stage Encrypted Exfiltration)**
Affected Assets	Subnet 192.168.1.0/24: Host IPs 192.168.1.7, 192.168.1.10, 192.168.1.3
Attack Vector	Abuse of Recursive Port 53 External DNS with Encoded High-Entropy Payloads
Incident Status	**CONTAINED & REMEDIATED - Malicious C2 Domains Sinkholed**

Section b. Detection and Analysis

Security telemetry was evaluated across five heuristic dimensions to calculate an explainable Risk Score (0-7): Domain Length (>50 chars), Subdomain Depth (>=2 subdomains), Shannon Entropy (>3.5 bits/char), Query Frequency (>=5 queries), and TXT Query Types. Live packet captures using Scapy and PCAP inspections revealed active Base64-encoded strings exfiltrating data chunks to *.example.com and *.exfil-c2.net.

Section c. Mitigation and Recovery

1. Host Isolation: Endpoints 192.168.1.7, 192.168.1.10, and 192.168.1.3 isolated onto a quarantine VLAN.

2. DNS Sinkholing: Perimeter DNS resolvers configured to sinkhole malicious adversary domains to 127.0.0.1.

3. Firewall Egress Filtering: Enforced perimeter rule blocking direct outbound UDP/TCP Port 53 except via internal DNS.

4. Endpoint Remediation: Quarantined malicious PowerShell/Python beacon processes and revoked Active Directory tokens.

5. Recovery Verification: Telemetry validated 0 subsequent anomalous TXT queries prior to reconnecting hosts.

Section d. Future Recommendations

- Preventive Controls: Implement DNS Response Policy Zones (RPZ) and Zero Trust Network Architecture (ZTNA).
- SOC Optimization: Ingest real-time Shannon entropy metrics directly into the centralized SIEM ingestion pipeline.
- Policy Enhancement: Restrict local DNS overrides and mandate centralized DNS over TLS (DoT) proxying.
- Security Training: Conduct targeted anti-phishing simulations for end users to eliminate initial access vectors.